

LEUKOS

The Journal of the Illuminating Engineering Society

ISSN: 1550-2724 (Print) 1550-2716 (Online) Journal homepage: www.tandfonline.com/journals/ulks20

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To cite this article: Georgios Tsiakiris , J. Rahm , U. Besenecker , M. Hedblom , S. Dincel , E. Persson & M. Johansson (25 Aug 2026): Assessing Lighting Scenarios and Environmental Experiences for Pedestrians in After-Dark Urban Forests, LEUKOS, DOI: 10.1080/15502724.2026.2711970

To link to this article: https://doi.org/10.1080/15502724.2026.2711970

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Published online: 25 Aug 2026.

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LEUKOS
https://doi.org/10.1080/15502724.2026.2711970

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Assessing Lighting Scenarios and Environmental Experiences for Pedestrians in After-Dark Urban Forests

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ABSTRACT

Urban forests are considered important for well-being and a habitat for animals year-round, including dark seasons. This study concerns pedestrians' appraisals of electric lighting scenarios of an urban forest path, with some of the scenarios being designed to decrease lighting influence on local wildlife. We explored how pedestrians experience walking in a local urban forest in three electric lighting scenarios: conventional LED light as reference lighting (RL); a modified luminaire with clear and amber lenses in lighting scenario 1 (LS1); and identical to LS1 but dimmed to 11% of LS1's lumen output in lighting scenario 2 (LS2). Participants, $N = 78$ in a between-subject design, engaged in structured walks and provided observer-based environmental assessments onsite. Participants' self-report assessments included environmental experiences and behavioral intentions. Significant differences between lighting scenarios were shown for visual accessibility (LS1 > LS2), prospect (RL > LS2), perceived safety (LS1 > LS2), perceived strength quality (PSQ) (RL > LS2, LS1 > LS2), perceived comfort quality (PCQ) (LS1 > RL, LS2 > RL), and restorative potential (RL > LS2). The findings indicate that people need a certain level of light to feel safe and psychologically restored. Further, the perceived comfort of the lighting appears to be greater for LS1 and LS2, featuring a luminaire with warm-white LEDs and a mix of clear and amber lenses, than for a conventional neutral-cool white LED scenario. Amber light with reduced short-wavelength radiation is currently assumed to have less impact on many non-human species; thus, the evaluated design may lower the ecological impact.

ARTICLE HISTORY
Received 13 February 2026
Revised 6 July 2026
Accepted 13 July 2026

KEYWORDS
Electric light; Non-human species; Pedestrian; Psychological restoration; Urban forest

1. Introduction

Urban forests are considered to offer great support to psychological well-being (Ode Sang et al. 2016) and to biodiversity (Croci et al. 2008). Urban and peri-urban forests (200–7500 m buffer around cities) are defined as tree-dominated ecosystems in and near human settlements (Lehto et al. 2022). These forests are important components of healthy and livable cities (UN 2019), and it would benefit public health if they were used year-round. Investigating how human health is promoted by urban forests can provide valuable information for planning and management (Kondo et al. 2018). In Sweden, forests are the most-used habitat for outdoor recreation, and 44% of all recreational activities in Sweden take place in urban forests (Lehto et al. 2022). So far, most research on the health benefits of urban forests has been conducted in the spring, summer, and fall seasons (Lehto et al. 2022; Lévesque-Vézina and Lapointe 2025). During the winter season, recreation is conducted closer to one's home, highlighting the importance of nearby urban forests (Lehto et al. 2022). However, in countries with limited daylight hours during the winter season, such as Sweden, recreational opportunities in urban forests during dark hours are limited. While dark natural environments may facilitate certain recreational activities, such as stargazing and night photography (Beeco et al. 2023), other activities are dependent on sufficient light for navigation. Electric lighting is often introduced to increase sight in public environments compatible with human needs, and while improving the conditions for humans, electric lighting often has negative effects on non-human species in these

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settings (Longcore and Rich 2004; Lewanzik et al. 2022; Rodrigo-Comino et al. 2023).

This research concerns pedestrians' appraisals of different scenarios of electric lighting in an urban forest path. To "quantify" the notion of compatible urban forests with people's needs, we investigated appraisals proposed to be important for walking after dark (Johansson et al. 2024; Tsiakiris et al. 2025), such as visual accessibility (Johansson et al. 2011), prospect, escape (Nasar 2008), perceived safety (Blöbaum and Hunecke 2005), cognitive vulnerability in a wildlife encounter (Johansson et al. 2012), perceived strength and comfort quality of outdoor lighting (Johansson et al. 2014), perceived restorative potential (Hartig et al. 1997) and wildlife restorative potential (Johansson et al. 2024a), along with the behavioral intentions to choose or avoid using a similar urban forest path in the future (Johansson et al. 2016).

Pedestrians' appraisals of daylight and after-dark conditions show that pole-mounted electric lighting installed along a pedestrian path can provide adequate levels of perceived visual accessibility, and support for prospect and escape, perceived safety, and psychological restoration in urban streetscapes (Rahm et al. 2024). The same effects of electric lighting have been shown in urban forest settings (Tsiakiris et al. 2025). However, such appraisals may differ depending on how the light is distributed (Bullough et al. 2020) and the correlated color temperature (CCT) of the light source. An experimental laboratory study showed that, in urban street lighting, the CCT can have significant effects on the visual perceptions (Hao et al. 2022), and the CCTs that are considered most preferable by laypeople for walking paths range between 3000–3500 K (Jin et al. 2015; Petrulis et al. 2018; Davidovic et al. 2019). Boyce et al. (2000) indicated that nighttime safety perception, in relation to daytime safety perception, was improved as the average illuminance in lit parking lots increased, but with relatively smaller improvements as the light level increased above 20 to 30 lux. Further, a laboratory study evaluating LED lighting in a light booth with different CCTs (2700, 4000, & 6500K) found statistically significant differences between the light sources on visual tests, with 2700K being rated the most comfortable, pleasant, and relaxing (Viénot et al. 2009).

Anthropogenic light impacts natural systems in a multifaceted way. As for non-human species, that animals and insects have distinct reactions (e.g., attraction or repulsion) to light is a well-known direct response (Jägerbrand and Spoelstra 2023), for example, negative responses to light among some bat species (Lewanzik et al. 2022) where light creates a barrier for them. There are no simple solutions that can lead to electric light mitigation, since light sources with different characteristics (e.g., spectrum, distribution, and strength) can evoke various responses from different species (Jägerbrand and Spoelstra 2023). Various studies have been conducted suggesting benefits of adjusting the spectral content of the light sources to amber-appearing lighting (Longcore 2023) (reduced blue-rich wavelengths range from 2700K down to 1800K). One approach for reducing the impact of electric lighting is to change the CCT, since amber-appearing lighting, having reduced short-wavelength when compared to neutral-cool white light, is proposed to be less negative after dark for most species and to contribute less to overall skyglow (Esposito and Radetsky 2023). Newer LED lighting technologies provide the adjustable technical possibilities for wildlife-adapted LED illumination applications (Dincel et al. 2025). The choice to use amber light and interpret it as less negative on wildlife is based on limited literature, as this topic of inquiry is still relatively recent. According to a recent overview by Gaston et al. (2025), they conclude that there are "too few studies to allowing amber light being a panacea for changing light in cities for animals." Yet, they claim that PC amber LED light is undoubtedly an improvement over PC-white LED. They suggest that there are other potential options that might be even better (Gaston et al. 2025). However, at the time this study was conducted, warm white and amber were selected due to the emerging practice to reduce specifically short-wavelength radiation in the after-dark hours outdoors (demonstrated to disrupt certain wildlife behaviors and increase light scattering into the night sky).

1.1. Lighting Scenarios

Considering both human vision and ecological impact as described above, two lighting scenarios were developed for the present study and were installed along an urban forest path. The scenarios

were intended to provide sufficient lighting for pedestrians while decreasing disturbances to wildlife. Compared to the existing lighting scenario on the path (reference lighting, RL), lighting scenario 1 (LS1) had a similar light output as RL but with a mix of clear and amber lenses on warm white LEDs. Lighting scenario 2 (LS2) was identical to LS1 but dimmed to 11% of LS1 lumen output to meet the criteria of the lighting class P4 for pedestrian routes in the outdoors (Swedish Transport Administration 2020). The P4 lighting criteria are suited for paths with minor nighttime use where the consideration of the local environment should be taken into account (Swedish Transport Administration 2020). The lighting scenarios are described in detail in section 2.3. LS1 and LS2 were installed on the same lampposts as the existing (RL) with LED light sources that were previously evaluated in a comparison of day and night conditions for pedestrians in urban forests (Tsiakiris et al. 2025).

1.2. Theoretical Background

Our study investigated human interaction with the physical environment—in our case, an urban forest path, in response to several different lighting scenarios—therefore, we applied the Human-Environment Interaction model (HEI model) (Küller 1991; Johansson 2006) as a conceptual model. The HEI model depicts the continuous interaction of people with their physical environment in a given setting, considering the activity with which the people engage in their social environment and the individuals' personal characteristics (e.g. gender, age). Since the activity in focus for our study was walking in an urban forest setting, we included the hierarchy of walking needs (Alfonzo 2005). It is a structure proposing certain environmental experiences as important to promote walking, i.e., feasibility—not applicable in this study since it relies on personal capabilities, and the willingness to walk the path was an inclusion criterion—accessibility, safety, comfort, and pleasurability of the urban environment.

Accessibility is operationalized as the perceived visual accessibility of the path (Johansson et al. 2011), which applies to the functional levels a pedestrian needs for orientation and obstacle detection (Rahm and Johansson 2021), along with prospect (perceived overview of the surroundings) and escape (perceived opportunities to escape the setting) (Rahm et al. 2021). When an individual perceives an environment as having a high level of entrapment—a similar construct to escape—lighting conditions become less important for the perception of safety, while prospect levels gain importance (Blöbaum and Hunecke 2005).

Safety is operationalized as perceived safety (Blöbaum and Hunecke 2005), along with the aforementioned prospect and escape (Boyce et al. 2000; Bullough et al. 2020). Moreover, safety in the present study is further operationalized as cognitive vulnerability in a wildlife encounter situation (inverted), comprising the perceived danger, the unpredictability of animal behavior, and a lack of control in a wildlife encounter (Johansson et al. 2024a).

The aspect of comfort is operationalized as the perceived outdoor lighting qualities, which consist of two dimensions: the perceived strength quality (PSQ) and perceived comfort quality (PCQ) (Johansson et al. 2014).

Pleasurability is operationalized as the restorative potential of the setting, referring to the extent to which an environment is perceived to help the individual in restoring their psychological resources (Hartig et al. 1997) and further operationalized as the wildlife restorative potential, namely, the perception of wildlife presence as supporting the potential for gaining psychological restoration in a given natural setting (Johansson et al. 2024b). Environmental experiences referring to wildlife were captured with regard to two aspects: 1) safety in the present study is further operationalized as cognitive vulnerability in a wildlife encounter situation (inverted), comprising the perceived danger, the unpredictability of animal behavior, and a lack of control of one's own reaction in a wildlife encounter, and 2) the wildlife restorative potential, namely, the perception of wildlife presence as supporting the potential for gaining psychological restoration in a given natural setting. In the questionnaire items, the animal was exemplified with roe deer, *Capreolus capreolus*.

Environmental appraisals of walking paths are associated with behavioral intentions to choose or

avoid walking the path in the future (Johansson et al. 2016), and therefore, we included these walking intentions in our conceptual model.

Our theoretical background (Fig. 1) is not only anthropocentric; it is also linked to the premise that human needs are not and should not be the only needs of value and rather aims toward an ecosystem well-being theory (Fleurbaey and Leppanen 2021). The presence of other than human species is not specific in the urban forest at hand (we do not register specific species in our study), but still our approach is aspired to be species neutral, meaning providing lighting for humans while decreasing the impact on a multitude of animal species when compared to a neutral white reference condition.

1.3. Study Aim

This study aims to assess pedestrian experience of three different lighting scenarios installed in an urban forest path: RL (existing reference lighting), LS1 (amber), and LS2 (amber and dimmed).

1.4. Research Questions (RQ1 – RQ2)

RQ1: Are there any differences between RL, LS1, and LS2 in pedestrian experience on visual accessibility, prospect, escape, perceived safety, cognitive vulnerability, perceived strength quality (PSQ), perceived comfort quality (PCQ), restorative potential, and wildlife restorative potential?

RQ2: Are there any differences between RL, LS1, and LS2 in the behavioral intention to choose and avoid a similar urban forest path depending on the lighting scenarios?

1.5. Hypotheses (H1 – H4)

Considering differences in the properties of the lighting scenarios resulting in a higher light output of RL and LS1 as compared to LS2, a cooler color temperature of RL as compared to LS1 and LS2, as well as more frequent reports of glare for RL than LS1 and LS2 (for further details, please refer to section 2.3), the following hypotheses are formulated:

Related to visual accessibility and safety (H1):

a. RL and LS1 will be appraised higher than LS2 on visual accessibility.
b. RL and LS1 will be appraised higher than LS2 on prospect.
c. In escape possibilities, RL, LS1, and LS2 may show differences.

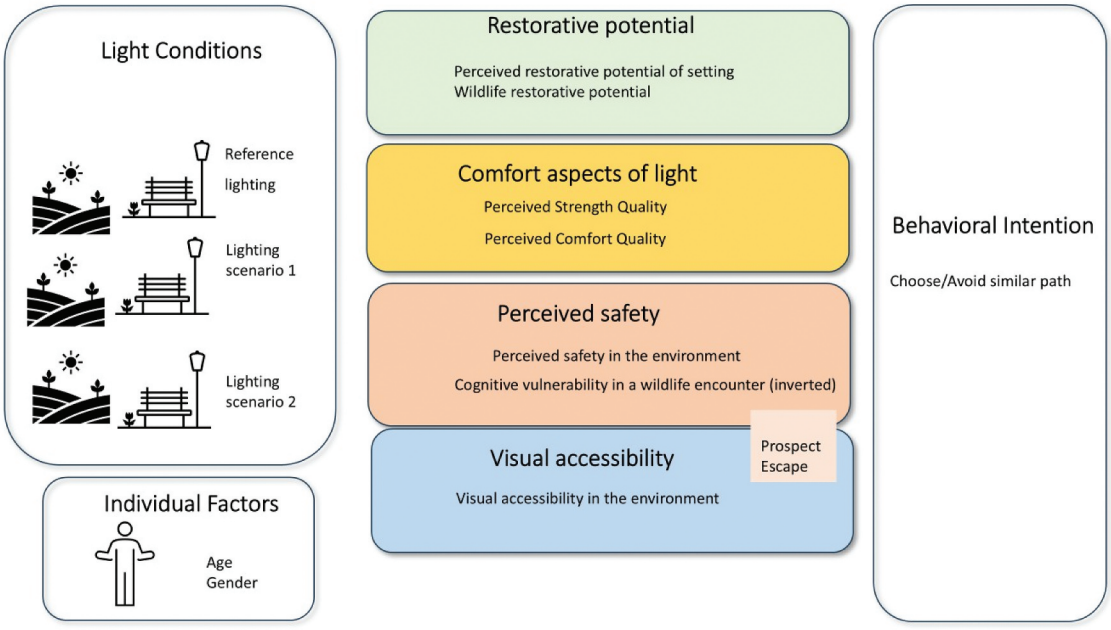


Fig. 1. Conceptual model and operationalizations of environmental experiences considered important for walking.

d. RL and LS1 will be appraised higher than LS2 on perceived safety.
e. RL and LS1 will be appraised lower than LS2 in cognitive vulnerability because of the better overview of the natural environment.

Related to perceived outdoor lighting qualities (H2):

a. RL and LS1 will be appraised higher than LS2 for PSQ based on the lower luminous flux.
b. LS1 and LS2 will be appraised higher than RL for PCQ based on the lower level of CCT.

Related to opportunities for psychological restoration (H3):

a. RL and LS1 will be appraised higher than LS2 in restorative potential.
b. As well as in wildlife restorative potential, RL and LS1 will be appraised higher than LS2.

H4: RL and LS1 will be appraised higher than LS2 in the behavioral intention to choose (H4a) and lower in the behavioral intention to avoid a similar path (H4b), because of the lower luminous flux and the differences in CCT.

2. Method

2.1. Setting

The study was conducted in the neighborhood of Sävja (10,000 inhabitants), located 6 km from the city center of Uppsala (167,000 inhabitants), Sweden. The neighborhood consists of a combination of detached and attached family housing, located adjacent to an urban forest. The study was located on a graveled urban forest path that was created for leisure for cyclists and pedestrians in a forest on the urban fringe. The participants walked 270 m on the path that was approx. 3 m wide and ran in parallel with housing units at a distance of 50–100 m. For about 25 m on both sides of the path, the forest was managed for safety reasons with shrubs in the understory cleared. The area is frequently used by people, and several smaller paths lead from the houses to the gravel path. The forest area is municipally owned, and since there is no production demand, the area is in its natural state. Part of the forest is adjacent to an N2000 area (nature protection areas) and a nature reserve (Swedish Environmental Protection Agency 2018).

2.2. Participants

Twenty-seven structured walks (see section 2.4) with observer-based environmental assessments were made outdoors on-site by means of established questionnaire batteries (Johansson et al. 2016). In total, the study involved 78 participants (men $n = 22$, women $n = 50$, and $n = 6$ did not specify) aged 23–82 years (mean age = 48 yrs, SD = 15 yrs), divided between the three electric lighting scenarios: RL, LS1, and LS2 (Table 1). All participants were residents in the broader study area.

The study had a between-subject design, meaning that participants passed only one lighting condition. As it was not possible to easily change the lighting scenarios and thereby have different presentation orders, the between-subject design avoided biases due to order effects. We balanced the three groups of participants as far as possible: all participants were local residents, the mean age, age distribution, and gender distribution were approximately the same in the three groups,

Table 1. Descriptive statistics of the participants per lighting scenario.

	Reference lighting	Lighting scenario 1	Lighting scenario 2	Total
Number of participants	25	25	28	78
Women	16	15	19	50
Men	7	9	6	22
Did not specify	2	1	3	6
Age range	29–82 yrs	23–73 yrs	26–78 yrs	23–82 yrs
M(SD)	47 (14)	47 (17)	51 (15)	48 (15)

along with women being the majority. The study design was taken into account, along with assumptions of effect sizes when estimating the sample sizes needed for the study. A post-hoc power analysis showed that different measures had a power ranging between .75 and .99.

2.3. Lighting Scenarios

Lamp posts (4.5 m high) installed by the Uppsala municipality were positioned at intervals of 28–32 m on one side along the entire path. For this study, the municipality changed the pole-top luminaires on 12 poles. Light output (in lumens), horizontal and vertical illuminance levels (in lux) per lighting scenario, is shown in Fig. 2.

RL consisted of pole-top luminaires (Prisma Light Elton) along the path with white LED light sources, widely used in outdoor Nordic environments (power: 18W, luminous flux: 2100 lumen, CCT: 3000 Kelvin). The average illuminance on the path during dark conditions (quarter moon) was 12 lux (Range: 45 lux to near-zero), and the uniformity was 0.07 (Dincel 2023). In a first analysis of user experiences (same sample as the present study), 40% of the pedestrians experienced glare when walking under RL, compared to only 24% for LS1 and 28,6% for LS2 (Dincel and Tsiakiris 2026) (see Fig. 3).

LS1 was designed to have the same lumen output as RL (2100 lumens) (Dincel 2025). Using the same lampposts as RL, new customized pole-top luminaires (Focus/Fox) were installed (described in Dincel 2025). Each luminaire (power: 22W, CCT: 2700 Kelvin source, 2300K (1800–2400K depending on angle after lensing) was equipped with two different kinds of lenses (LEDiL). Clear lenses in the center of the luminaire directed light directly at the path, and amber lenses at the edges of the luminaire cast light with reduced short-wavelength radiation (lower CCT) at the areas next to the path. The average illuminance on the path during dark

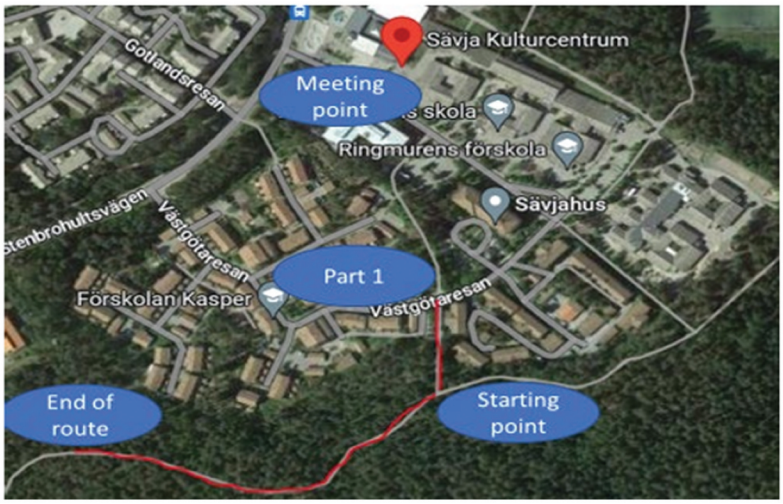


Fig. 2. Map of the structured walk with designated places of meeting point, part 1, start and end of route. Source: Lantmäteriet, 1:8500.



(a) (b) (c)

Fig. 3. Reference lighting (a), lighting scenario 1 (b) and lighting scenario 2 (c).

conditions (new moon) was 12 lux, the lowest was 0.47 lux (Range: 55.5 lux to near-zero), and the uniformity was 0.008 (Dincel 2023).

LS2 used the same customized luminaire as LS1; however, the light output—luminous flux—was dimmed to 11% of the original lumen output, producing 231 lumens per fixture. The average illuminance on the path during dark conditions (full moon) was 6 lux (Range: 22,77 lux to near zero), and the uniformity was 0.05 (Dincel 2023). The lighting design of LS1 and LS2 achieved an amber-appearing light in the periphery (at high angles) (CCT = 1800K), and warm-white-appearing light in the center (CCT = 2400K) projected onto the path (Dincel 2025). The prototype luminaire as supplied had a CCT of 2300K (1800–2400K depending on angle), and the light output was rated at 2100 lumens (Dincel 2025), which means that there was a reduction in CCT from the existing lighting (3000K) with increasingly amber-appearing light (less short-wavelength content) at high angles. This was done to minimize the more harmful short-wavelength light spreading into the forest areas. Forty percent of the pedestrians experienced glare for RL, compared to only 24% for LS1 and 28,6% for LS2 (Dincel and Tsiakiris 2026) (see Fig. 3).

Table 2 summarizes the average light levels for each experimental condition, measured under dry and overcast conditions during new and third-quarter moon phases.

During the walks, selected participants (one per group) were wearing portable light sensors (LYS button) (LYS Technologies 2021). The LYS buttons were pinned on the participants' jackets in a front-facing direction. The direction of the LYS button thus depended on how the participants positioned themselves. For example, one date and time of the field studies (November 15, 2023 between 17:30 and 18:30) was selected to illustrate the average CCT that would illuminate people and their faces over the length of the walk in LS1. According to data logs from LYS button sensors, the average CCT recorded in the field study was 2474K (see Fig. 4).

Illuminance uniformity was measured on two grids between light poles 2 and 3 and between light poles 8 and 9 (see Fig. 5). The light measurement grid was developed and used for both baseline and post-installation light measurements. The following was considered: establishing a horizontal grid of measurement points on the site surface; ensuring uniform spacing between measurement points; having spacings between measurement points less than one-half the pole

Table 2. Luminaire light output in lumens, horizontal illuminance and vertical illuminance in the field in lux under dry, overcast sky conditions (1 × 2 m grid between poles, based on one measurement session per scenario, as shown in Table 3). See Dincel (2023) and Dincel (2025) for additional information on measurement procedures and values.

Lighting scenario	Light output per pole top luminaire (in lm)	Horizontal illuminance M(SD) (in lux) and CCT M(SD) (in k)	Vertical illuminance M(SD) (in lux) and CCT M(SD) (in k)
Reference	2100	10.9 (2.3) 3186 (35.1)	5.5 (1.3) 3664 (117.4)
lighting Lighting	2100	11.7 (2.8) 2784 (75.8)	5.8 (2.1) 3266 (180)
Scenario 1 Lighting	231	5.8 (0.8) 3837 (97.6)	5.2 (0.8) 4405 (211.8)

Table 3. Dates and times of the structured walks and the light measures.

Lighting scenario	Dates of walk /light measure (MM/DD/YYYY)	Time of the walks	Daylight condition	Moon phases
Reference lighting 18	11/24/2022	−17.30 & 19.00	Astronomical twilight/ Night	All conditions
Walk	01/26/2023	16:52–19:27	Astronomical twilight/Night	First quarter
	Measure 12/07/2022			
Lighting scenario 1	11/14/2023	−17:30	Astronomical twilight/Night	Third quarter – new
Walk	11/29/2023		Night	moon – first quarter
Lighting scenario 2	Measure 11/27/2023	18:04–19:28	Night	New moon
Walk	03/11/2024	19:00	Nautical/Astronomical	Full moon – third quarter
	03/21/2024			
	12/03/2024	17:30	Night	Full moon
	Measure 01/13/2025	18:12–19:12	Night	Full moon

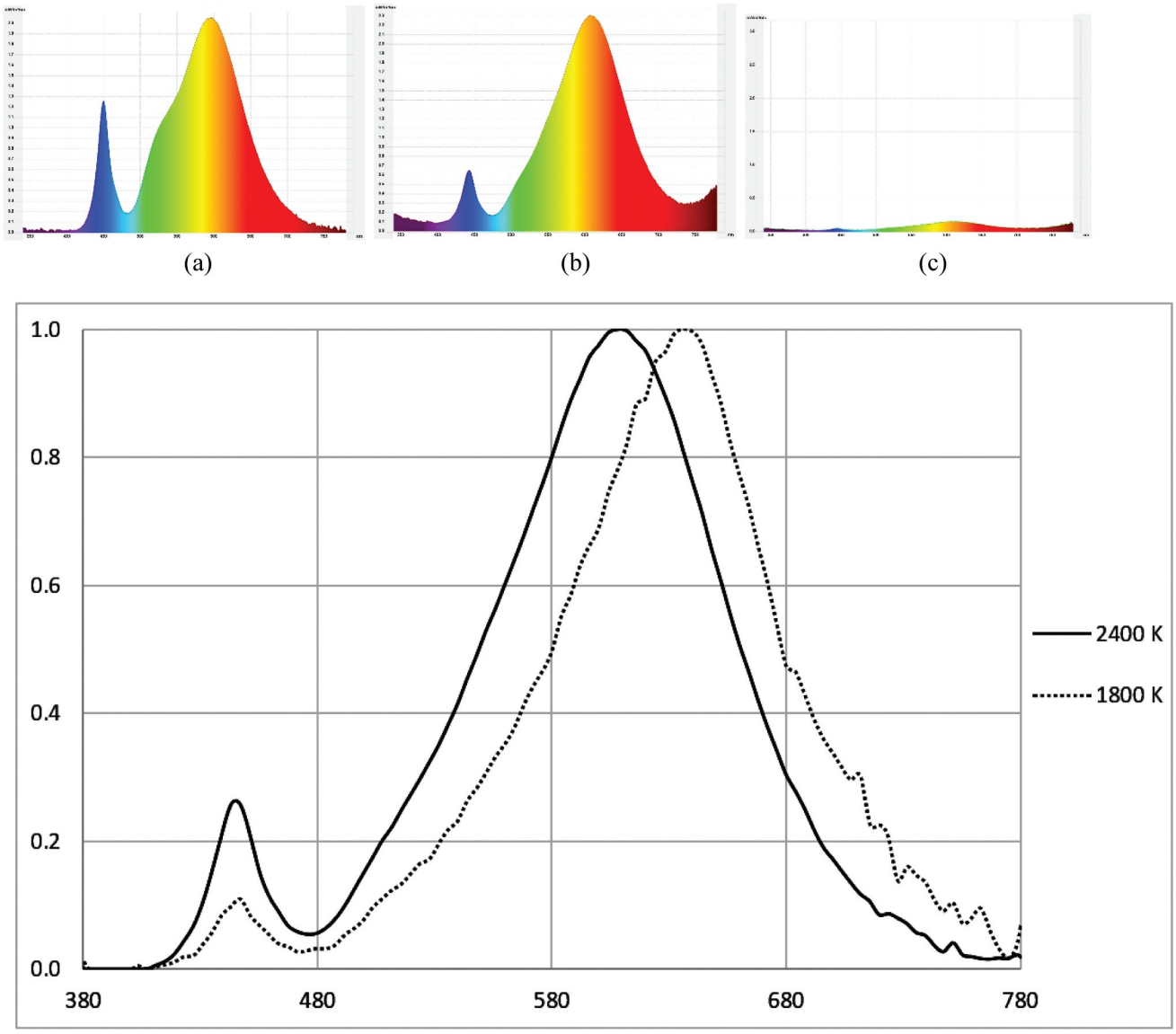


Fig. 4. SPD diagrams of reference lighting (a), lighting scenario 1 (b), and lighting scenario 2 (c), along with the clear and amber spectra (below). SPD diagrams: Seren Dincel.

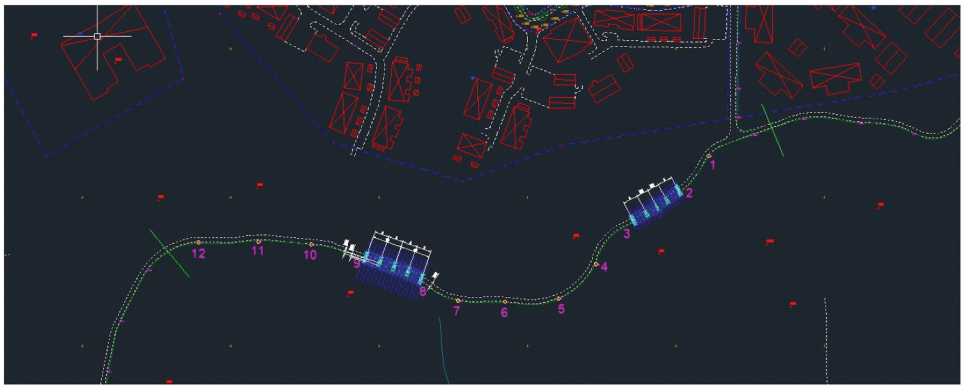


Fig. 5. Areas selected for the light measurements, full detailed measurements between poles 2 and 3, and control measurements between poles 6 and 7 (from Dincel 2023).

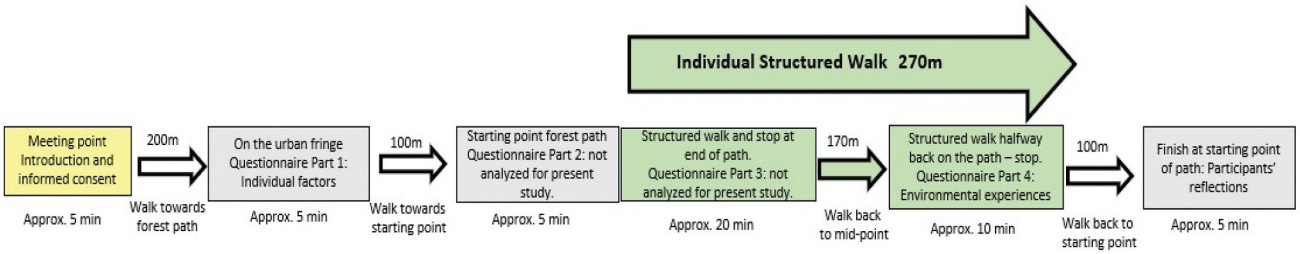


Fig. 6. Flowchart of the structured walks.

height (pole height 4.5 meters); and identifying a vertical plane representative of the use and lighting in the area (Dincel 2023).

2.4. Procedure

The recruitment process for the structured walks took place on-site (Sävja), through advertising on social media and through personal networks. The participants who signed up received written information about the study beforehand through e-mails and verbally on-site. They were informed that they could withdraw from the study at any time without explanation. Written informed consent was obtained from all participants before the start of the structured walk (see Ethics below).

At the time of the walks, the researcher met the participants in small groups (maximum six people) at the local community center 300 m from the path and walked together toward the designated path. At a point approximately 100 m from the start of the route, the participants completed the first part of the questionnaire. When part 1 was completed, the researcher walked with the participants to the starting point of the route, where they completed part 2. Afterward, participants walked the path individually, leaving a distance of approximately 60 m from one another. At the end of the designated route, a sign on light pole number 10 stated, "The route ends here." The participants completed part 3 by the sign, without the intervention of the researcher, who remained at a distance. This was done to let the participants experience walking the path and nature without being disturbed. The participants were then led by the researcher back to the midpoint of the path, where they completed part 4. Lastly, after returning to the starting point, the participants were asked to reflect verbally upon their assessments. The overall procedure lasted 40–50 minutes for each group (see Fig. 6).

The structured walks that were conducted during the winter season for the RL condition were conducted between November 2022 and January 2023. The structured walks for LS1 were conducted in November 2023 and for LS2 in March and December 2024.

The structured walks took place during astronomical twilight for all scenarios to ensure that the darkness conditions were similar. The weather varied slightly during each scenario, but temperatures were in the range of –10 to 3 degrees Celsius, which is not unusual for the study locations. Although a big challenge due to the fact that snow comes early at this latitude in Sweden, we excluded walks that took place in snow conditions from the analyses due to the truly different experience of the surroundings. We avoided walks in heavy rain.

2.5. Measurements

2.5.1. Questionnaire

The participants completed a 16-page questionnaire, which consisted of four parts: Part 4, analyzed for this paper, covered the assessments of visual accessibility, prospect, escape, perceived safety, cognitive vulnerability, PSQ, PCQ, restorative potential, wildlife restorative potential, and behavioral intention to choose and to avoid the path. Parts 1–3 were not analyzed for this paper: value orientation and trait connectedness to nature (Part 1) and valence, arousal, perceived restoration, and state connectedness to nature (Parts 2 and 3). Measurements included in the study are described below.

2.5.2. Measurements of environmental appraisals

To assess whether the three different light scenarios affected the pedestrians' experience of the lit environment, several measurements were used.

Visual accessibility was measured by seven statements: "I could have detected an object on the ground; I could have read street signs; I could have recognized people's faces; I could have detected cracks and irregularities in the path; Discern trees and bushes along the path; Could see what grew on the soil across the path; In general, I could see well." A five-point response scale was used (1 = *absolutely not* to 5 = *yes, absolutely*), and the responses were averaged into an index (αRL = .70, αLS1 = .84, αLS2 = .88).

Prospect was assessed by the statement "I have good overview of the surroundings along the path," and escape was assessed by the statement "I could easily escape the path in the event of danger." Both items were rated using five-point scales (1 = *absolutely not* to 5 = *yes, absolutely*) (Rahm et al. 2021).

Perceived safety (Blöbaum and Hunecke 2005) was assessed by five statements: "I would go a long way around to avoid this place (reversed); I feel uneasy at this place (reversed); I would make haste to get away from this place (reversed); I have an unpleasant feeling at this place (reversed); I would walk along this path unaccompanied." The participants rated each statement using a five-point scale (1 = *absolutely not* to 5 = *yes, absolutely*). Responses were averaged into an index (αRL = .81, αLS1 = .92, αLS2 = .94).

Cognitive vulnerability was assessed by six items: "I think that the movement of (animal) is impossible to understand in advance; I believe that if I came close to an (animal) I would be harmed; I believe that I would be able to deal effectively with an (animal) by myself if encountered (reversed); I find an (animal) to be predictable in their movements (reversed); I do not believe an (animal) could be dangerous to me (reversed); If an (animal) came near me, I would probably not feel in control." Responses were given on five point-scales (1 = *absolutely not* to 5 = *yes, absolutely*) and averaged into an index (αRL = .73, αLS1 = .84, αLS2 = .79) (Johansson et al. 2024a).

PSQ and PCQ were assessed with the Perceived Outdoor Lighting Quality scale (POLQ) (Johansson et al. 2014) using seven-point semantic differential scales. PSQ: "Clear – Drab (reversed), Strong – Weak (reversed), Unfocused – Focused, Subdued – Brilliant, Dark – Light" (αRL = .75, αLS1 = .66, αLS2 = .77) and PCQ: "Mild – Sharp (reversed), Hard – Soft, Warm – Cool (reversed), Glaring – Shaded, Natural – Unnatural (reversed)" (αRL = .76, αLS1 = .71, αLS2 = .72). Perceived restorative potential (Hartig et al. 1997): respondents indicated the perceived restorative potential of the setting by the statements: "There is much to explore and discover here; Spending time here gives me a good break from my day-to-day routine; My attention is drawn to many interesting things; The setting has fascinating qualities." Responses were given on five-point scales (1 = *absolutely not* to 5 = *yes, absolutely*) and averaged into an index (αRL = .70, αLS1 = .84, αLS2 = .90).

Wildlife restorative potential was measured with two items on encountering a wild animal (deer); the participant would assess "As dangerous and threatening for me (reversed); I can be engaged and delighted by watching the animal's activity and movement" (Johansson et al. 2024b). Responses were indicated on five-point scales (1 = *absolutely not* to 5 = *yes, absolutely*) and averaged into an index.

The behavioral intention was assessed by two statements for the question "What would you do if this route was at the right direction to one of your destinations in the next week?" ("Specifically choose walking along a similar path" and "Specifically avoid walking along a similar path"), using a five-point scale (1 = *absolutely not* to 5 = *yes, absolutely*) (Johansson et al. 2016).

2.6. Ethics

All participants had good command of the Swedish language. Informed consent was obtained from all participants at the meeting point. Personal information was anonymized to retain the privacy of the participants. They all received a cinema ticket, approximately 10 EUR in value, after participating. The study has ethical approval with reference number 2022–03857-01 from the Swedish Ethical Review Authority.

2.7. Statistics and Analyses

The statistical analyses were performed with IBM SPSS 29.0.0. Descriptive statistics were obtained (median, mean, and standard deviation) (Table 4). Parametric (ANOVA) and non-parametric (Kruskal–Wallis) tests were conducted based on the data distribution (skewness and kurtosis per lighting scenario) for each variable, along with effect size. Level of significance was set at .05.

3. Results

3.1. Visual Accessibility

Visual accessibility was provided in RL (mRL = 4.16) and LS1 (mLS1 = 4.31) and, to a lesser extent, LS2 (mLS2 = 3.81). Kruskal–Wallis tests show statistically significant differences in visual accessibility between the different light scenarios H(2) = 7.14, p = .028 η^2H = .07. Posthoc tests showed a statistically significant difference in pairwise comparisons between LS1 and LS2 (z = 2.63; p = .025, r = .36) partly confirming H1a.

3.2. Prospect

Prospect was to some extent provided in existing lighting (mRL = 3.38) and LS1 (mLS1 = 3.29) but to a limited extent in LS2 (mLS2 = 2.59). One-way ANOVA showed statistically significant differences in prospect between the different lighting scenarios, F (2, 75) = 4.38, p = .016, η^2p = .105. Post-hoc tests showed a difference between RL and LS2 (p = .028, r = .38, partly confirming H1b.

Table 4. Median, mean values, and standard deviation (SD) for each investigated variable per condition.

	Reference lighting (RL) Median/Mean/SD	Lighting scenario 1 (LS1) Median/Mean/SD	Lighting scenario 2 (LS2) Median/Mean/SD	Statistics for the light condition	(Significant) Post hoc Group comparison
Visual accessibility	4.29 / 4.16 / .51	4.43 / 4.31 / .61	3.93 / 3.81 / .76	**H (2) = 7.14, p = .028 η^2 = .07** p	**LS1 with LS2 (H(2) = 2.63, p = .025, r = .36)**
Prospect	4.00 / 3.38/ .95	4.00 / 3.29 / 1.24	2.00 / 2.59 / .99	**(F (2, 75) = 4.38, p = .016, η^2= .105) p**	**RL with LS2 (p = .028, r = .38**
Escape	3.33 / 3.33 / 1.14	3.00 / 3.00 / 1.23	2.00 / 2.74 / 1.24	F (2, 75) = 1.605, p = .208, **η^2p** = .04	–
Perceived safety	4.20 / 4.11 / .90	4.80 / 4.30 / .87	3.59 / 3.39 / 1.25	**H(2) = 9.38, p = .009, η2 H = .10**	**LS1 with LS2 z = 2.98, p = .009, r = .41**
Cognitive vulnerability	1.83 / 2.02 / .59	2.04 / 2.04 / .77	2.00 / 2.08 / .64	H(2) = .067, p = .967, **η^2**H = .000	–
PSQ	4.40 / 4.53 / 1.02	4.20 / 4.47 / .92	3.10 / 3.29 / .90	**(F (2, 75) = 14.71, p < .001, η^2= .282)**	**RL with LS2 (p < .001, r = .55) LS1 with LS2 (p < .001, r = .55)**
PCQ	3.80 / 4.13 / .81	5.00 / 4.85 / .90	5.20 / 5.05 / .77	**(F (2, 75) = 8.87, p <.001, η^2= .191)**	**RL with LS1 (p = .009, r = .39) RL with LS2 (p < .001, r = .51)**
Restorative potential	4.23 / 4.23 / .46	3.80 / 3.79 / .75	3.80 / 3.42 / .95	**H(2) = 11.64, p = .003 η^2 = .13**	**RL with LS2 (z = 3.37, p = .002, r = .46)**
Wildlife restorative potential	4.63 / 4.63 / .46	4.50 / 4.15 / 1.14	4.08 / 4.17 / .83	(H(2) = 3.91, p = .142, η2 H = .03)	–
Behavioral intention (choose)	4.00 / 4.28 / .74	4.26 / 4.26 / .83	4.00 / 3.61 / 1.34	(H(2) = 4.00, p = .135, η2 H = .03	–
Behavioral intention (avoid)	2.00 / 2.00 / 1.19	1.96 / 1.96 / 1.14	2.00 / 2.36 / 1.31	H(2) = 2.40 p = .301 **η^2** = .01 p	–

3.3. Escape Possibilities

Escape possibilities were around 3 in RL (mRL = 3.33), LS1 (mLS1 = 3.00), and LS2 (mLS2 = 2.74). A one-way ANOVA showed no statistically significant differences, $F(2, 75) = 1.605$, $p = .208$, $\eta^2 p = .04$, thus not confirming H1c.

3.4. Perceived Safety and Vulnerability

Perceived safety was provided in RL (mRL = 4.11) and LS1 (mLS1 = 4.30) and, to a lesser extent, in LS2 (mLS2 = 3.39). Kruskal–Wallis tests showed statistically significant differences between the different lighting scenarios: $H(2) = 9.38$, $p = .009$, $\eta^2 H = .10$. Post-hoc tests showed a significant difference in pairwise comparisons between LS1 and LS2 ($z = 2.98$, $p = .009$, $r = .41$), thus partly confirming H1d.

Cognitive vulnerability was relatively low in RL (mRL = 2.01), LS1 (mLS1 = 2.03), and LS2 (mLS2 = 2.08). Kruskal–Wallis tests showed no statistically significant differences between the different lighting scenarios: $H(2) = .067$, $p = .967$, $\eta^2 H = .000$; thus, H1e could not be confirmed.

3.5. PSQ and PCQ

Perceived strength quality of the light was provided for RL (mRL = 4.53), LS1 (mLS1 = 4.47), and, to a lesser extent, LS2 (mLS2 = 3.29). ANOVA showed statistically significant differences between the lighting scenarios, $F(2, 75) = 14.71$, $p < .001$, $\eta^2 p = .282$. Post-hoc tests showed statistically significant differences between RL and LS2 ($p < .001$, $r = .55$) and between LS1 and LS2 ($p < .001$, $r = .55$), thus fully confirming H2a.

Perceived comfort quality was provided for RL (mRL = 4.13), LS1 (mLS1 = 4.85), and LS2 (mLS2 = 5.05). ANOVA showed statistically significant differences between the lighting scenarios, $F(2, 75) = 8.87$, $p < .001$, $\eta^2 p = .191$. Post-hoc tests showed statistically significant differences between RL and LS1 ($p = .009$, $r = .39$) and between RL and LS2 ($p < .001$, $r = .51$), thus fully confirming H2b.

3.6. Restorative Potential of the Setting and Wildlife Restorative Potential

Restorative potential was provided for RL (mRL = 4.23), LS1 (mLS1 = 3.79), and, to a lesser extent, LS2 (mLS2 = 3.41). Kruskal–Wallis tests showed statistically significant differences between the different lighting scenarios ($H(2) = 11.64$, $p = .003$, $\eta^2 H = .13$). Post-hoc tests showed a statistically significant difference between RL and LS2 ($z = 3.37$, $p = .002$, $r = .46$), thus partly confirming H3f.

The wildlife restorative potential was provided for RL (mRL = 4.63), LS1 (mLS1 = 4.15), and LS2 (mLS2 = 4.17). Kruskal–Wallis tests showed no statistically significant differences between the different lighting scenarios ($H(2) = 3.91$, $p = .142$, $\eta^2 H = .03$), thus not confirming H3g.

3.7. Behavioral Intention

The behavioral intention to choose a similar path was assessed for RL (mRL = 4.28), LS1 (mLS1 = 4.26), and LS2 (mLS2 = 3.61). Kruskal–Wallis tests showed no statistically significant differences between the lighting scenarios ($H(2) = 4.00$, $p = .135$, $\eta^2 H = .03$), thus not confirming H4a.

The behavioral intention to avoid the path was assessed for RL (mRL = 2.00), LS1 (mLS1 = 1.96), and LS2 (mLS2 = 2.36). Kruskal–Wallis tests showed no statistically significant differences between the lighting scenarios ($H(2) = 2.40$, $p = .301$, $\eta^2 H = .01$), thus not confirming H4b.

3.8. Participants' Reflections

In the verbal reflections after the walks, there was a variation during dark hours in reference to lighting, with some participants stating that they would walk there only in company of others and others that they would not hesitate to walk alone, although perceived safety was a recurring theme (Tsiakiris et al. 2025).

After LS1 walks, participants in their personal reflections found that the activity at hand—walking—fit well with the lighting scenarios. The lighting was rated as strong, creating contrasts with the sides and between the light poles, and was

perceived as warmer than the previous conventional lighting (RL). Participants thought LS1 was softer and more natural than RL. There were different opinions on whether LS1 was glary or not. However, participants mentioned that they could not see deep into the forest. Others said that they did not feel secure and that the lighting created a "wall" to the forest. It was also mentioned that in terms of safety, there was a difference between walking in a group and walking alone. Participants mentioned that the path was wide, which helped with wildlife encounters, but also that not being able to see into the forest would not help if a "running moose" were to appear.

After LS2 walks, participants stated that the lighting was soft, a bit dark, and nice, although focused on specific points and uneven. They would like to see better between the light poles, but admitted that there were no hard contrasts. Moreover, participants stated that they would like to see better toward both sides of the path, which was explicitly in relation to other people, not animal presence. Some participants considered the lighting environment scary; however, that came from people who had not walked in that specific area before. It was mentioned that walking in a group was different than walking alone, and the necessity for lighting accommodating human needs was understood. Participants replied that in terms of wildlife encounters, it would depend on which type of animal they would meet, and that wildlife reactions on their behalf would be independent of lighting. LS2 lighting was deemed appropriate to enjoy nature, and the noises of the environment were brought up. Participants mentioned that they were afraid of the darkness and that the path could evoke emotions different than daylight. Some participants mentioned that a different type of light would not make walking more appealing, and even if it was brighter, that would not ease their negative emotions. However, other participants stated that the lighting was appropriate for the path and the specific rural area and that it generated relaxed emotions. LS2 was appraised as softer lighting and a good "compromise" to cater to both human and wildlife considerations. All verbal reflections gave a contextual understanding of the pedestrian experience.

4. Discussion

Electric lighting in urban forests may improve conditions for human physical activity, recreation, and psychological restoration, while at the same time causing light pollution disturbing non-human species. This is especially cumbersome during the Nordic winter season, when it is dark during the day. This study aimed to tackle this problem by evaluating two lighting scenarios (RL, LS1 and LS2) along an urban forest path. The scenarios were customized to reduce short-wavelength light spilling into the adjacent forest and thereby limit the impact on other species inhabiting the urban forest (Dincel 2025). The results show that LS1 (amber light in the periphery and white center) performed better than LS2 (amber light in the periphery and white center + light output reduced to 11%) in visual accessibility, perceived safety, and perceived strength quality (PSQ). The study participants' appraisals of the path were quite similar for LS1 and the reference lighting (RL). The RL performed better than LS2 in visual accessibility, PSQ, and restorative potential. However, for the perceived comfort quality (PCQ), both LS1 and LS2 were appraised as providing a better comfort quality of the light than RL.

The average illuminance levels measured on the path after LS2 was implemented corresponded to the P4 class of pedestrian lighting appropriate in rural areas such as the study area (with full moon during measures adding to the light levels), and although it was appraised lower in most of the aforementioned variables, it did perform better in PCQ than RL. This result is in line with previous research suggesting that brightness is important to visual accessibility and perceived safety (Fotios et al. 2015; Bullough et al. 2020) but not necessarily that the light is perceived as being comfortable. Our findings contribute to a better understanding of visual accessibility and perceived safety for a more sustainable lighting design, based on real-life pedestrian experiences of different lighting scenarios (with each having distinctive but comparable characteristics) in the same setting.

The only statistically significant difference between RL and LS1 was in PCQ, with LS1

performing better. Hence, both customized lighting scenarios performed better than RL in PCQ.

In a previous study where four different lighting applications were tested in the field, assessments of perceived comfort quality of the light were associated with the preference for each light application (Patching et al. 2017). In yet another study using a full-scale laboratory to evaluate pedestrian lighting, the lighting application that had the highest mean illuminance was rated the highest on perceived strength quality and the lowest on perceived comfort quality (Rahm and Johansson 2018). According to Rahm and Johansson (2018), the results for perceived comfort quality could potentially stem from the differences in correlated color temperature (hard-soft and warm-cool). It may also be explained by a reduction of the perceived experience of glare when looking at the sky during the field study (Dincel and Tsiakiris 2026). Light can be transmitted away from the source, and even a single strong light can be a focal element of the landscape, resulting in sight difficulties and glare perception in the night sky (Jägerbrand and Spoelstra 2023).

As for the perceived restorative potential, RL was assessed as higher than LS2, indicating that people need visual accessibility levels and visual information to perceive the environment as restorative. Visual accessibility levels are crucial for recreational purposes when it comes to walking an urban forest path, but the dimmed lighting could assist in dark-based recreational activities, so one cannot draw holistic conclusions.

In wildlife restorative potential, there were no statistically significant differences between the lighting scenarios, presumably because of the lack of wildlife presence during the walks. At the same time, all three scenarios showed relatively high wildlife restorative potential (mean values above 4), which means that all lighting scenarios helped the participants feel that in a wildlife encounter, they would be able to control the situation, again indicating that visual information would be sufficient.

There were no statistically significant differences between the lighting scenarios in the participants' behavioral intentions to specifically choose or avoid a similar urban forest path.

This result may indicate that light conditions may not be the most critical factor in participants' walking decisions among people who are already familiar with the area (Wilkie and Trotter 2022). Instead, the social environment, along with fear of crime, could also be a reason (Nikunen and Korpela 2012; Abraham et al. 2025).

The finding that LS1 resulted in higher perceived comfort quality of the light could be used as an argument to apply it permanently on the path, since the lighting scenario was also assessed to have a sufficient perceived strength quality. Moreover, LS1 would be a compromise for the species balance since amber light has also been suggested for the benefit of certain non-human species, although further studies are needed to identify the broader effects of such lighting (Esposito and Radetsky 2023; Jägerbrand and Spoelstra 2023).

The present study lowered CCT below the 3000K mark, and there was no difference between the RL and LS1 (except for PCQ in favor of LS1), indicating that people's needs can be met with warmer-appearing light than expected. The warm lighting chosen for LS1 and LS2 still had a broad spectral distribution, enabling recognition of different surface colors illuminated by the light; some amber sources (e.g., low-pressure sodium, narrow-band amber LEDs) do not render colors well and should be avoided in general lighting applications where colors need to be distinguishable. In this regard, there is a need to find a balance between dark and illuminated environments (Katabaro et al. 2022), especially in settings such as urban forests where the landscape is shared between humans, animals, and plants. People may need lighting to facilitate wayfinding (Davoudian 2017), but at the same time, animals also get cues from light, e.g., moonlight for their behavior; essentially, they need natural dark and light cues to navigate or trigger other behavioral and physiological responses, as they evolved according to such cues in their habitat. Prospect was assessed higher for RL than LS2, indicating that less dimming (<89% dimming) could be more functional for supporting an overview of the surroundings. An increase from 89% dimming could be more functional in terms of having an

overview of the surroundings. In escape possibilities, there were no significant differences, indicating that other aspects of the physical environment—apart from lighting—may have a greater impact on perceived possibilities for escape.

Traditionally, outdoor recreational activities, such as forest walks, have been common in the Nordic countries (Ankre et al. 2016; Fredman et al. 2023), establishing the significance of our study in terms of human recreation and psychological restoration. Testing lighting interventions that aim to be less intrusive to other species can help develop practical suggestions to decision-makers for more balanced lighting in urban forest paths. Our results suggest that it would be worthwhile for future studies to look into "in between" lighting scenarios, namely below LS1 and above LS2, to investigate how people would experience such conditions.

The design of the structured walk has to be mentioned; participants knew that one researcher and other participants were present, probably making the setting feel safer than when one walks all alone, as mentioned in the participants' reflections. Still, real-life walks in an urban forest path during (winter) after dark hours are close to the actual experience.

The theoretical model of this study argues that people's walking experience in an urban forest path can be dissected into several aspects. In every aspect of our theoretical model (visual accessibility, safety, comfort, pleasurability), there was a variable on which lighting had a significant effect, verifying the different walking needs of an individual who visits an urban forest.

Electric outdoor lighting constitutes a source of artificial light at night (ALAN) and light pollution. According to the DarkSky organization, light pollution refers to human-made alteration of outdoor light levels from those occurring naturally, manifesting as light trespass, skyglow, glare, and clutter (DarkSky International 2025). Electric lighting can disrupt how animals function in an urban forest (Haddock et al. 2019), and that shows the necessity for customization or limitation of ALAN in established outdoor lighting characteristics (Lewanzik et al. 2022; Dincel et al. 2025) and practices that take into consideration both human and other species. The present study tests and suggests alterations to lower light spill/trespass by directing the light on the path with a distinct but smooth transition into the forest, altering the spectral power distribution (SPD) and CCT to reduce short-wavelength radiation, and reducing the overall light output (Lewanzik et al. 2022).

5. Conclusion

This study examined how three different lighting scenarios along an urban forest walking path can balance human needs—such as visual accessibility, perceived safety, lighting comfort, and restorative potential—while reducing artificial light at night and light spill into surrounding habitats in an urban forest. By evaluating two customized lighting scenarios compared to a municipal reference lighting condition, the findings provide practical insights into how changes in lighting characteristics can influence pedestrians' experiences without necessarily increasing the ecological impact. The results indicate that brightness, color temperature, and light distribution affect human environmental experiences in different ways. LS1, combining clear lenses in the center of the luminaire directing light directly at the path and amber lenses at the edges of the luminaire, casting light with lower CCT in the areas next to the path, emerged as a promising compromise. It performed similarly to the reference lighting in terms of lumen output while being appraised higher in perceived comfort quality. This suggests that people's requirements for functional lighting that supports perceptions of safety in urban forest paths can be met with warmer lighting. Lighting with the quality characteristics of LS1 therefore represents a solution that supports human walking experiences while potentially reducing negative impacts on non-human species through lower spill of short-wavelength radiation. LS2, a dimmed version of the customized lighting, was generally rated lower in terms of visual accessibility, prospect, safety, PSQ, and restorative potential, yet it received favorable evaluations for comfort quality.

This highlights that reduced brightness does not necessarily result in negative overall experiences; however, a minimum level of visual information is important for people to perceive the urban forest

environments as restorative, particularly for an activity such as walking. Taken together, the findings support the idea that urban forest lighting can consider a combination of warm-white and amber lighting when designing strategies to balance pedestrian usage and experience with ecological impact. It should be noted that ecological impact was not evaluated in this study; neither the impact of the luminaire on overall use of electricity and resources nor the specific impact of light radiation on local species were tested and addressed. This study focused on the human experience to assess whether low-output yellowish-amber lighting solutions would compromise human usability, with the outcome providing information that is useful for both application and further research.

Disclosure Statement

No potential conflict of interest was reported by the author(s).

Author contributions

CRediT: **Georgios Tsiakiris:** Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Project administration, Visualization, Writing – original draft, Writing – review & editing; **J. Rahm:** Conceptualization, Data curation, Formal analysis, Methodology, Supervision, Visualization, Writing – review & editing; **U. Besenecker:** Conceptualization, Funding acquisition, Methodology, Writing – review & editing; **M. Hedblom:** Supervision, Writing – review & editing; **S. Dincel:** Visualization, Writing – review & editing; **E. Persson:** Visualization, Writing – review & editing; **M. Johansson:** Conceptualization, Data curation, Formal analysis, Funding acquisition, Methodology, Project administration, Supervision, Writing – review & editing.

Funding

This work was supported by Nordforsk via the Research Council Formas grant number [Dnr: 2021-00384]; Swedish Energy Agency grant number [Dnr: P2021-00024] (under Nordark Research Consortium, https://nordark.org). Georgios Tsiakiris was partly funded by the Lund University Agenda 2030 Graduate School.

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